



**TRIOS CYBER**

**LABORATORY PRACTICE 17**

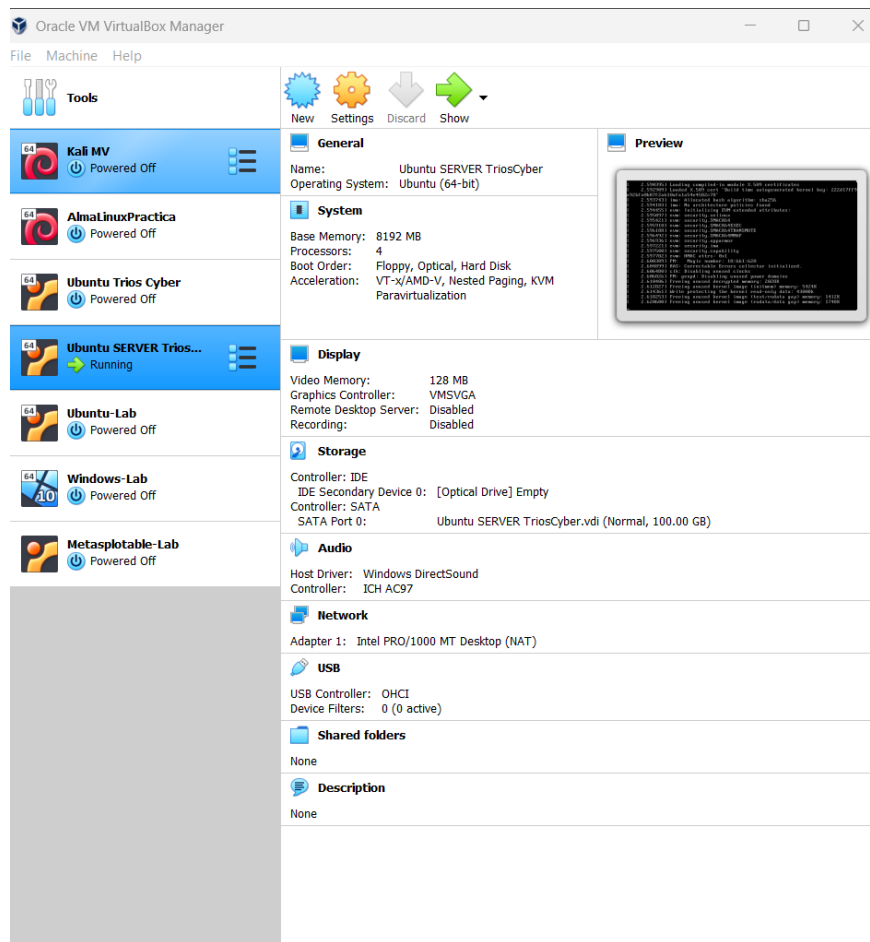
Wazuh All-in-One Installation and Dashboard Verification

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# 1. Wazuh Server Virtual Machine Resource Configuration

The dedicated Ubuntu Server virtual machine was prepared in Oracle VirtualBox for the Wazuh all-in-one deployment. Before the final installation, the VM configuration was adjusted to provide 8192 MB of memory and 4 virtual processors while retaining the 100 GB virtual disk and the NAT network adapter. The server was then started and used as the centralized Wazuh platform for the laboratory.



## 2. Installation Workflow Preparation

The deployment was performed on the Ubuntu Server host using the Wazuh installation assistant in all-in-one mode. This workflow installs the Wazuh indexer, Wazuh manager, Filebeat, and Wazuh dashboard on the same server. The installation was allowed to complete before service validation and browser access were tested.

### 3. Wazuh All-in-One Installation Result

The installation assistant completed the indexer post-install configuration, initialized the indexer cluster, installed and started the Wazuh manager, installed and started Filebeat, and completed the Wazuh dashboard installation. The dashboard web application was then initialized successfully and the assistant reported that the installation had finished.

The generated administrator password has been intentionally redacted from the evidence included in this report. The administrator username is not treated as a secret, but the password must not be exposed in submitted evidence.

```

matthew@trioscyber: ~
19/09/2026 16:23:51 INFO: Wazuh indexer post-install configuration finished.
19/09/2026 16:23:51 INFO: Starting service wazuh-indexer.
19/09/2026 16:24:09 INFO: wazuh-indexer service started.
19/09/2026 16:24:09 INFO: Initializing Wazuh indexer cluster security settings.
19/09/2026 16:24:12 INFO: Wazuh indexer cluster security configuration initialized.
19/09/2026 16:24:12 INFO: Wazuh indexer cluster initialized.
19/09/2026 16:24:12 INFO: --- Wazuh server ---
19/09/2026 16:24:12 INFO: Starting the Wazuh manager installation.
19/09/2026 16:25:42 INFO: Wazuh manager installation finished.
19/09/2026 16:25:42 INFO: Wazuh manager vulnerability detection configuration finished.
19/09/2026 16:25:42 INFO: Starting service wazuh-manager.
19/09/2026 16:25:52 INFO: wazuh-manager service started.
19/09/2026 16:25:52 INFO: Starting Filebeat installation.
19/09/2026 16:27:10 INFO: Filebeat installation finished.
19/09/2026 16:27:17 INFO: Filebeat post-install configuration finished.
19/09/2026 16:27:17 INFO: Starting service filebeat.
19/09/2026 16:27:18 INFO: filebeat service started.
19/09/2026 16:27:18 INFO: --- Wazuh dashboard ---
19/09/2026 16:27:18 INFO: Starting Wazuh dashboard installation.
19/09/2026 16:49:05 INFO: Wazuh dashboard installation finished.
19/09/2026 16:49:06 INFO: Wazuh dashboard post-install configuration finished.
19/09/2026 16:49:06 INFO: Starting service wazuh-dashboard.
19/09/2026 16:49:07 INFO: wazuh-dashboard service started.
19/09/2026 16:49:08 INFO: Updating the internal users.
19/09/2026 16:49:15 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
19/09/2026 16:49:30 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
19/09/2026 16:50:26 INFO: Initializing Wazuh dashboard web application.
19/09/2026 16:50:28 INFO: Wazuh dashboard web application initialized.
19/09/2026 16:50:28 INFO: --- Summary ---
19/09/2026 16:50:28 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password:
19/09/2026 16:50:28 INFO: Installation finished.
matthew@trioscyber:~$

```

## 4. Core Service Verification

After installation, the four main services were checked individually with `systemctl is-active`. The Wazuh manager, Wazuh indexer, Wazuh dashboard, and Filebeat each returned active, confirming that the all-in-one components were running at the time of validation.

```
matthew@trioscyber:~$ echo "=== WAZUH MANAGER ==="
sudo systemctl is-active wazuh-manager
echo "=== WAZUH INDEXER ==="
sudo systemctl is-active wazuh-indexer
echo "=== WAZUH DASHBOARD ==="
sudo systemctl is-active wazuh-dashboard
echo "=== FILEBEAT ==="
sudo systemctl is-active filebeat
=== WAZUH MANAGER ===
[sudo] password for matthew:
active
=== WAZUH INDEXER ===
active
=== WAZUH DASHBOARD ===
active
=== FILEBEAT ===
active
```

## 5. SSH Tunnel for Local Browser Access

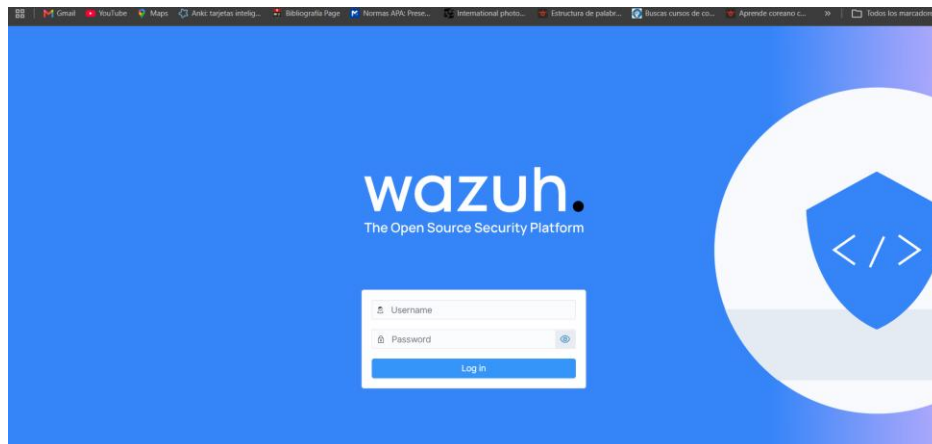
Because the Ubuntu Server was running behind the VirtualBox NAT configuration, a local SSH tunnel was created from the Windows host. Local TCP port 8443 was forwarded through the SSH session to 127.0.0.1:443 on the Wazuh server, allowing the HTTPS dashboard to be reached from the host browser without exposing the service externally.

```
PS C:\WINDOWS\system32> ssh -N -L 8443:127.0.0.1:443 -p 2222 matthew@127.0.0.1
matthew@127.0.0.1's password:
```

| Tunnel element           | Observed value                      |
|--------------------------|-------------------------------------|
| Windows browser endpoint | https://127.0.0.1:8443              |
| SSH host connection      | matthew@127.0.0.1 on host port 2222 |
| Forwarded destination    | 127.0.0.1:443 on the Ubuntu Server  |

## 6. Wazuh Login Interface Access

The local tunnel was kept open and the Wazuh web interface was opened from the Windows browser. The HTTPS login page loaded successfully, confirming that the dashboard service was reachable through the forwarding path. Browser chrome and personal bookmarks were cropped from the report evidence because they are unrelated to the laboratory task.

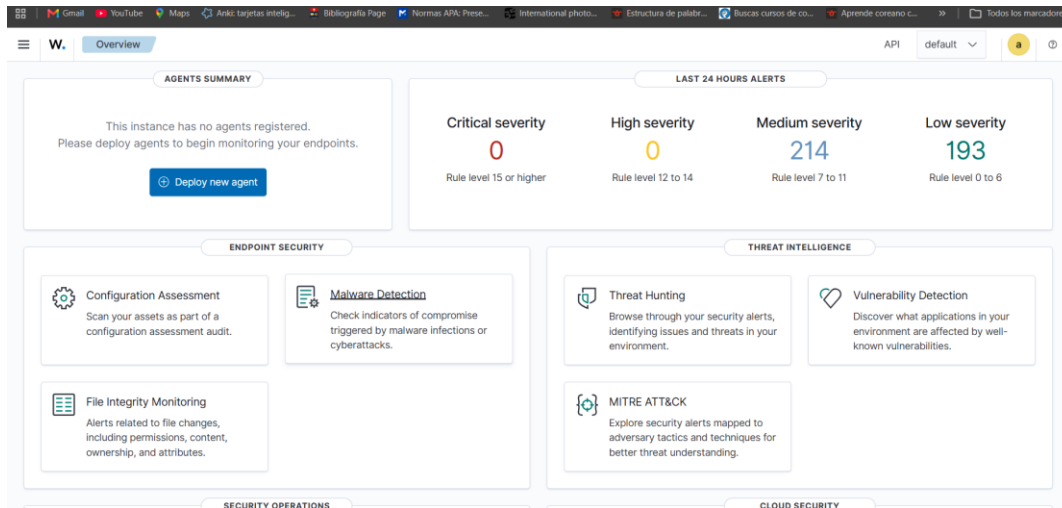


## 7. Dashboard Authentication

The administrator account generated by the installation workflow was used to authenticate to the Wazuh interface. Credentials are not reproduced in the report. Successful authentication led to the Wazuh Overview page shown in the following evidence.

## 8. Wazuh Dashboard Access and Validation

The Wazuh Overview page loaded successfully after authentication. The interface displayed the Agents Summary, Last 24 Hours Alerts, Endpoint Security, and Threat Intelligence areas, confirming functional browser access to the installed dashboard. At this stage, the instance reported that no endpoint agents were registered; agent deployment was outside the scope of Day 17.



## 9. Deployment Summary

| Component / Check | Observed Evidence                                        | Result                  |
|-------------------|----------------------------------------------------------|-------------------------|
| Ubuntu Server VM  | 8192 MB RAM, 4 vCPU, 100 GB virtual disk, NAT adapter    | Ready                   |
| Wazuh manager     | systemctl is-active wazuh-manager                        | Active                  |
| Wazuh indexer     | systemctl is-active wazuh-indexer                        | Active                  |
| Wazuh dashboard   | systemctl is-active wazuh-dashboard                      | Active                  |
| Filebeat          | systemctl is-active filebeat                             | Active                  |
| Browser access    | Wazuh login and Overview loaded through local SSH tunnel | Verified                |
| Agent state       | No endpoint agents registered at the time of validation  | Not required for Day 17 |

## 10. Conclusions

The Wazuh all-in-one deployment was successfully completed on the Ubuntu Server virtual machine. The manager, indexer, dashboard, and Filebeat services were confirmed as active, and an SSH tunnel provided secure local access from the Windows host to the HTTPS dashboard. The Wazuh login interface and Overview page were opened successfully in the browser, satisfying the installation, service-verification, and dashboard-access requirements of the laboratory.